Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.
Code: 100312
Subject: Mathematics III (PDE, Probability & Statistics)
Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): [2 x 7 = 14]

(a) The order of the PDE obtained by eliminating f from $Z = f(x^2 + y^2)$ is
(i) First (ii) Second (iii) Third (iv) Fourth

(b) Who was the first person to develop the heat equation?
(i) Galileo Galilee (ii) Jseph Fourier
(iii) Daniel Gabriel (iv) none of these

(c) The linear partial differential equation of order one is known as
(i) Lagrange's linear PDE (ii) Cauchy's linear PDE
(iii) Charpits linear PDE (iv) none of the above

(d) Poission distribution is
(i) Discrete (ii) Continuous (iii) Normal (iv) None of these

(e) Two unbiased coins are tossed simultaneously. The probability of getting less than 3 tails is
(i) 0 (ii) 1 (iii) 3/4 (iv) 1/2

(f) The total area under the normal curve above the x-axis is
(i) 0.5 (ii) 1 (iii) 2 (iv) 1.5

(g) The median for the series: 4, 6, 9, 4, 7, 8, 10 is
(i) 4 (ii) 7 (iii) 6 (iv) 8

(h) The relation between mean, median and mode is
(i) Mode =3Median – 2 Mean (ii) Mode = 2Median – 3 Mean
(iii) Mean = 3 Median – 2 Mode (iv) none of these

(i) If the regression coefficients are 0.8 & 0.2, what would be the value of coefficient of correlation?
(i) 0.6 (ii) 0.5 (iii) 0.4 (iv) 0.3

(j) The expectation of the number of heads in 15 tosses of a coin is
(i) 15 (ii) 30 (iii) 7.5 (iv) 8.5

Q.2 (a) Form partial differential equation by eliminating the function f from [7]

$$Z = e^{ax+by} f(ax - by).$$

(b) Solve the linear partial differential equation [7]

$$\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x\,\partial y} + \frac{\partial^2 z}{\partial y^2} = Sin(2x + 3y).$$

Q.3 (a) Obtain the general solution of heat flow equation $k\left(\frac{\partial^2 u}{\partial x^2}\right) = \frac{\partial u}{\partial t}$ by the method of separation of variables. [7]

(b) Show that a general solution of wave equation $c^2\left(\frac{\partial^2 \varphi}{\partial x^2}\right) = \frac{\partial^2 \varphi}{\partial t^2}$ is $\varphi = f(x + ct) + g(x - ct)$. [7]

Q.4 (a) Obtain solution of Laplace's Equation in cylindrical polar coordinates. [7]
(b) Determine whether the following equation is hyperbolic, parabolic, and elliptic? [7]

$$x\frac{\partial^2 u}{\partial t^2} + t\frac{\partial^2 u}{\partial x\,\partial t} + \frac{\partial^2 z}{\partial t^2} = 0.$$

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Q.5 (a) If X is a random variable such that E[X] = 3 and E[X²] =13, use the Chebyshev's inequality to determine the lower bound for P[-2 <X <8]. [7]

(b) Two integers are selected at random from 1 to 11. If the sum is even, find the probability that both the numbered are odd. [7]

Q.6 (a) State and prove Baye's theorem. [10]

(b) Find the mean number of heads in three tosses of a coin. [4]

Q.7 (a) The first four moments of a distribution about x = 2 are 1, 2.5, 5.5 and 16. Calculate the first four moments about the mean. [7]

(b) The following table gives the number of aircraft accidents that occurred during the various days of the week. Find whether the accidents are uniformly distributed over the week. (The tabulated value of Chi-square at 5% level for 6 degree of freedom is 12.59). [7]

Days	Sun.	Mon.	Tue.	Wed.	Thu	Fri	Sat	Total
No. of accidents	14	16	8	12	11	9	14	84

Q.8 (a) A random sample of 5 college students is selected and their grades in Mathematics & Statistics are found to be: [7]

Mathematics	85	60	73	40	90
Statistics	93	75	65	50	80

Calculate Spearman's rank coefficient.

(b) By the method of least square, find the straight line that best fits the following data: [7]

x	1	2	3	4	5
y	14	27	40	55	68

Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Binomial distribution

(b) Poisson Distribution

(c) Normal Distribution

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